

Model
$$X_t = \beta_0 + \beta_1 t + \underbrace{\gamma_t}_{\text{noise}}$$

Two estimators:

(1)
$$Z_t = X_t - X_{t-1}$$

$$\bar{Z}$$

(1) vs (2)

(2) Perform OLS to estimate the slope β_1

Conclusion: (2) is better

For (1), recall $Z_t = X_t - X_{t-1}$

$$\bar{Z} = \frac{Z_2 + Z_3 + \dots + Z_n}{n-1}$$

$$= \frac{(X_2 - X_1) + (X_3 - X_2) + (X_4 - X_3) + \dots + (X_n - X_{n-1})}{n-1}$$

$$= \frac{1}{n-1} (X_n - X_1)$$

just involves
2 samples.

- Repeated differencing

X_t : original time series.

$$Y_t = X_t - X_{t-1}$$

$$\begin{aligned} Z_t = Y_t - Y_{t-1} &= (X_t - X_{t-1}) - (X_{t-1} - X_{t-2}) \\ &= X_t - 2X_{t-1} + X_{t-2} \end{aligned}$$

$$\begin{aligned} W_t = Z_t - Z_{t-1} &= (X_t - 2X_{t-1} + X_{t-2}) - (X_{t-1} - 2X_{t-2} + X_{t-3}) \\ &= X_t - 3X_{t-1} + 3X_{t-2} - X_{t-3}. \end{aligned}$$

- Differencing & polynomial

$$X_t = a_n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0$$

$$\begin{aligned} X_t - X_{t-1} &= (a_n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0) - \\ &\quad (a_n (t-1)^n + a_{n-1} (t-1)^{n-1} + \dots + a_1 (t-1) + a_0) \end{aligned}$$

$$t^n - (t-1)^n = \text{polynomial of deg } n-1$$